

Scattering by a Central Potential; the Hamiltonian

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Scattering by a central potential

$$\begin{aligned} \text{Incident energy :} & \quad E = \frac{m}{2} v_0^2, \\ \text{Incident velocity :} & \quad v_0 = \sqrt{2E/m}, \\ \text{Angular momentum :} & \quad \ell = s m v_0 = s \sqrt{2mE}, \quad s = \frac{\ell}{\sqrt{2mE}}. \end{aligned}$$

$$d^3r = r^2 dr \underbrace{d\varphi \sin \theta}_{d\Omega}, \quad \text{scattered ring solid angle: } d\Omega = 2\pi \sin \theta d\theta.$$

The azimuthal angle φ is irrelevant by symmetry, so the deflection is controlled by $s(\theta, E)$: the impact parameter and the scattering angle are in one-to-one correspondence, $s \longleftrightarrow \theta$.

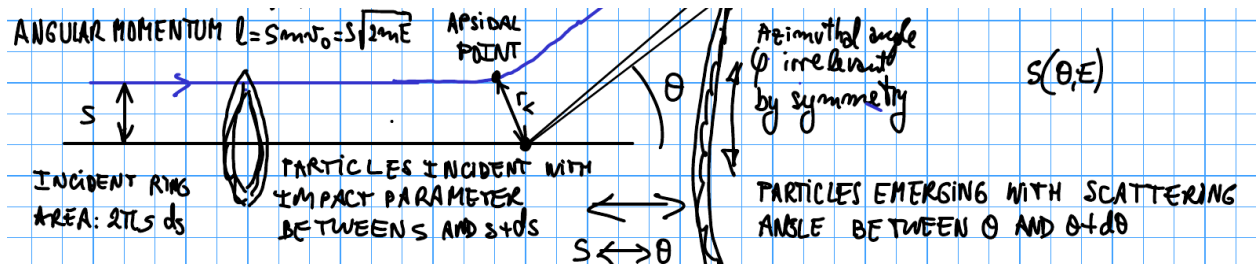


Figure 1: Particles incident in the ring of area $2\pi s ds$ emerge into the ring of scattering angles between θ and $\theta + d\theta$; $r_{<}$ is the distance of closest approach, reached at the apsidal point.

Incident intensity; differential cross section

$$\begin{aligned} I &= \frac{\# \text{ particles incident per unit time}}{\text{incident beam cross-section area}}, \\ \sigma(\Omega) d\Omega &= \frac{\# \text{ particles scattered into } d\Omega \text{ per unit time}}{\text{incident intensity}}. \end{aligned}$$

$$\begin{aligned} I(2\pi s) |ds| &= \# \text{ particles going through the ring between } s \text{ and } s + ds \\ &= \# \text{ particles emerging with angle between } \theta \text{ and } \theta + d\theta \\ &= I \sigma(\theta) 2\pi \sin \theta |d\theta|. \end{aligned}$$

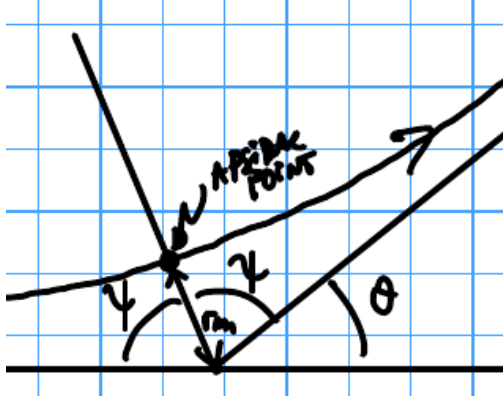
Therefore

$$\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right|$$

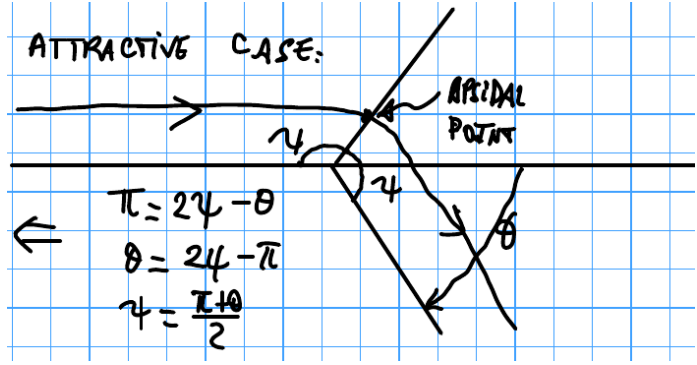
define also the total cross section:

$$\sigma_T = \int d\Omega \sigma(\Omega)$$

Repulsive and attractive geometry



Repulsive case



Attractive case

Figure 2: The apsidal angle ψ and the scattering angle θ in the two cases.

REPULSIVE CASE	ATTRACTIVE CASE
$\pi = \theta + 2\psi$	$\pi = 2\psi - \theta$
$\theta = \pi - 2\psi$	$\theta = 2\psi - \pi$
$\psi = \frac{\pi - \theta}{2}$	$\psi = \frac{\pi + \theta}{2}$

In all cases:

$$\theta = |\pi - 2\psi|$$

Computing $s(\theta, E)$ (universal method)

$$E = \frac{m}{2} \dot{r}^2 + V(r) + \frac{\ell^2}{2mr^2}, \quad \dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{\ell^2}{2mr^2} \right) \right]^{1/2}, \quad \dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}.$$

$$\begin{aligned} \ell = mr^2 \dot{\theta} \Rightarrow \frac{d\theta}{dt} = \dot{\theta} = \frac{\ell}{mr^2} \Rightarrow d\theta = \dot{\theta} dt = \frac{\ell}{mr^2} dt = \frac{\ell}{mr^2} \frac{dr}{\left[\frac{2}{m} \left(E - V(r) - \frac{\ell^2}{2mr^2} \right) \right]^{1/2}} \\ = \frac{\ell}{m} \sqrt{\frac{m}{2E}} \frac{r^{-2} dr}{\left[1 - \frac{V(r)}{E} - \underbrace{\frac{\ell^2}{2mE}}_{=s^2} \frac{1}{r^2} \right]^{1/2}}. \end{aligned}$$

Integrating from the apsidal point out to infinity gives the apsidal angle ψ ;¹ the substitution $u = 1/r$ is then made.²

$$\psi = \int_0^\psi d\psi = \int_{\theta_m}^\pi d\theta = \int_{r_m}^\infty \frac{s r^{-2} dr}{[1 - V(r) - s^2/r^2]^{1/2}}, \quad u = \frac{1}{r}, \quad du = -r^{-2} dr, \quad u_m = \frac{1}{r_m}$$

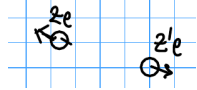
$$\Rightarrow \theta(s) = \left| \pi - 2 \int_0^{u_m} \frac{s du}{\left[1 - \frac{V(1/u)}{E} - s^2 u^2\right]^{1/2}} \right|$$

From here, compute

$$\left| \frac{ds}{d\theta} \right| = \frac{1}{\left| \frac{d\theta}{ds} \right|} \quad \text{and use it to get} \quad \sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right|.$$

Caveat 1. Up to here, results are valid for any central potential $V(r)$.

Coulomb scattering

Force $f = \frac{ZZ'e^2}{r^2},$  $f = -\frac{k}{r^2}, \quad V = -\frac{k}{r},$

$$k = -ZZ'e^2.$$

$E > 0$ hyperbolic

$E = 0$ parabolic

$E < 0$ elliptic

Only for $V = -\frac{k}{r}$!!!

Scattering: $E > 0$

$$\frac{1}{r} = C[1 + \varepsilon \cos(\tilde{\theta} - \tilde{\theta}')], \quad C = \frac{mk}{\ell^2} = -\frac{m}{\ell^2} ZZ'e^2.$$

$$\text{Eccentricity } \varepsilon = \left[1 + \frac{2E\ell^2}{mk^2}\right]^{1/2} = \left[1 + \frac{2E\ell^2}{m(ZZ'e^2)^2}\right]^{1/2} = \left[1 + \left(\frac{2sE}{ZZ'e^2}\right)^2\right]^{1/2} > 1,$$

$$\ell = smv_0 = s\sqrt{2mE} \rightarrow \frac{2\ell^2 E}{m} = \frac{s^2 2^2 \cancel{m} E^2}{\cancel{m}} = (2sE)^2.$$

Two situations:

i) Repulsive case $ZZ' > 0 \Rightarrow k = -ZZ'e^2 < 0;$

ii) Attractive case $ZZ' < 0 \Rightarrow k = -ZZ'e^2 > 0$ (also gravity, $k = Gm_1m_2$).

¹As written in the source, the integrand of the r -integral below carries $V(r)$ rather than $V(r)/E$: the division by E is omitted here although the boxed result on the same line does carry $V(1/u)/E$. Reproduced as written.

²The substitution is written in the source as $u = 1/r$, $du = r^{-2} dr$; the correct relation is $du = -r^{-2} dr$. The missing sign is absorbed by the interchange of the limits of integration. Reproduced as written.

i) Repulsive case: $k < 0$, $C = mk/\ell^2 < 0$

$$\begin{aligned} \frac{1}{r} = C[1 + \varepsilon \cos(\tilde{\theta} - \tilde{\theta}')] > 0 &\iff 1 + \varepsilon \cos(\tilde{\theta} - \tilde{\theta}') < 0 \\ &\iff \varepsilon \cos(\tilde{\theta} - \tilde{\theta}') < -1 \iff \cos(\tilde{\theta} - \tilde{\theta}') < -\frac{1}{\varepsilon}. \end{aligned}$$

We need the cosine to be negative and larger in magnitude than $1/\varepsilon$. In this case, $\tilde{\theta} = \tilde{\theta}'$ is not a physical point, because

$$\cos(\tilde{\theta}' - \tilde{\theta}') = \cos(0) = 1 \implies \underbrace{\frac{1}{r} = C[1 + \varepsilon]}_{\text{UNPHYSICAL}} < 0.$$

So $\tilde{\theta} = \tilde{\theta}'$ is in the middle of the inaccessible region of the hyperbola, and $\tilde{\theta} = \tilde{\theta}' \pm \pi$ will be the periapsis, so we choose $\tilde{\theta}' = \pi$ and we get

$$\cos(\tilde{\theta} - \tilde{\theta}') = \cos(\tilde{\theta} - \pi) = \cos \tilde{\theta} \underbrace{\cos \pi}_{-1} + \sin \tilde{\theta} \underbrace{\sin \pi}_0 = -\cos \tilde{\theta}.$$

$$\begin{aligned} \frac{1}{r} = \left(-\frac{mk}{\ell^2}\right) [-1 - \varepsilon \cos(\tilde{\theta} - \tilde{\theta}')] &= \frac{mZZ'e^2}{\ell^2} [\varepsilon \cos \tilde{\theta} - 1], & \text{maximum at } \tilde{\theta} = 0 \implies \\ \text{minimum } r : \frac{1}{r_m} &= \frac{mZZ'e^2}{\ell^2} [\varepsilon - 1]. \end{aligned}$$

So the angle at the apsidal point is $\tilde{\theta} = 0$. We now need

$$\begin{aligned} \psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}(\frac{1}{r} \rightarrow 0), & \quad \left[\text{in the repulsive case } \theta = \pi - 2\psi \implies \psi = \frac{\pi - \theta}{2} \right] \\ 0 = \varepsilon \cos \psi - 1 &\iff \frac{1}{\varepsilon} = \cos \psi = \cos\left(\frac{\pi - \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} + \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = \sin \frac{\theta}{2}. \end{aligned}$$

ii) Attractive case: $k > 0$, $C = km/\ell^2 > 0$

Take $\tilde{\theta}' = 0$:

$$\begin{aligned} \tilde{\theta} = \tilde{\theta}' = 0 \implies \frac{1}{r} = C[1 + \varepsilon \cos(0)] &= C[1 + \varepsilon], & \text{max at } \tilde{\theta} = 0 \\ \text{minimum } r : \frac{1}{r_m} &= C[1 + \varepsilon]. \end{aligned}$$

The apsidal point is also at $\tilde{\theta} = 0$. We now need

$$\begin{aligned} \psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}(\frac{1}{r} \rightarrow 0), & \quad \left[\text{in the attractive case } \theta = 2\psi - \pi \implies \psi = \frac{\pi + \theta}{2} \right] \\ 0 = 1 + \varepsilon \cos \psi &\iff -\frac{1}{\varepsilon} = \cos \psi = \cos\left(\frac{\pi + \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} - \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = -\sin \frac{\theta}{2}. \end{aligned}$$

Combining both cases

$$\cot^2 \frac{\theta}{2} = \frac{1}{\tan^2 \frac{\theta}{2}} = \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1 - \sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1}{\sin^2 \frac{\theta}{2}} - 1 = \varepsilon^2 - 1 = \left(\frac{2sE}{ZZ'e^2} \right)^2$$

$$\implies \boxed{s = s(\theta, E) = \left| \frac{ZZ'e^2}{2E} \right| \cot \frac{\theta}{2}}$$

$$\frac{ds}{d\theta} = \left| \frac{ZZ'e^2}{2E} \right| \frac{d}{d\theta} \left(\frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} \right) = \left| \frac{ZZ'e^2}{2E} \right| \left(\frac{1}{2} \left(\frac{-\sin \frac{\theta}{2}}{\sin \frac{\theta}{2}} \right) - \frac{1}{2} \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) = - \left| \frac{ZZ'e^2}{4E} \right| \frac{1}{\sin^2 \frac{\theta}{2}},$$

$$\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right| = \frac{1}{2} \left(\frac{ZZ'e^2}{2E} \right)^2 \frac{1}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} \frac{1}{\sin^2 \frac{\theta}{2}}.$$

$$\boxed{\sigma(\theta) = \frac{1}{4} \left(\frac{ZZ'e^2}{2E} \right)^2 \csc^4 \left(\frac{\theta}{2} \right)}$$

Rutherford scattering cross section
(for Coulomb forces)

Remarks:

- 1) The cross section is the same for the repulsive and attractive cases, since it is proportional to $(zz')^2$.
- 2) It diverges like E^{-2} at low energies.
- 3) It diverges like θ^{-4} (VERY STRONG!!!) for small angles θ .
- 4) This classical result is exactly the same as the result obtained in Quantum Mechanics.

Total scattering cross section

$$\sigma_T = \int \sigma(\Omega) d\Omega, \quad \text{in this case } \sigma(\theta) = \text{const} \frac{1}{\sin^4 \left(\frac{\theta}{2} \right)}.$$

$$d\Omega = 2\pi \sin \theta d\theta = 4\pi \sin \frac{\theta}{2} \cos \frac{\theta}{2} d\theta$$

$$\sigma_T = \frac{4\pi}{4} \left(\frac{ZZ'e^2}{2E} \right)^2 \int_0^\pi \frac{\sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\sin^4 \frac{\theta}{2}} d\theta, \quad \text{range: } 0 < \theta \leq \pi.$$

Near $\theta = 0$ the integrand behaves as

$$\sim \int_0^\delta \frac{d\theta}{\left(\frac{\theta}{2} \right)^3} = 8 \int_0^\delta \frac{d\theta}{\theta^3} = 8 \left. \frac{\theta^{-2}}{(-2)} \right|_0^\delta = +\infty \quad \text{STRONGLY DIVERGENT.}$$

NOTATION:

Goldstein

Others

$$\sigma(\Omega) \longleftrightarrow \frac{d\sigma}{d\Omega} \quad \text{differential scattering cross section}$$

$$\sigma_T \longleftrightarrow \sigma \quad \text{total scattering cross section}$$

Classical $\sigma_T = +\infty$ for any long range potential.
 Quantum $\sigma_T = +\infty$ for some long range potentials $V(r)$ that do not decay fast enough at $r \rightarrow \infty$.

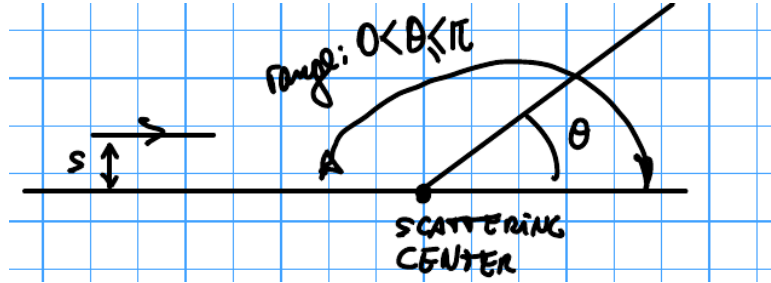


Figure 3: Deflection about the scattering centre; the accessible range is $0 < \theta \leq \pi$.

Caveat

1) In a dense conducting material:

$$V = -\frac{k}{r} \longrightarrow V^{\text{SCREEN}}(r) = -\frac{k}{r} e^{-r/\lambda} \quad \begin{array}{l} \text{screened Coulomb} \\ \text{potential ("Yukawa")} \end{array}$$

(In non-conducting materials $V^{\text{SCREEN}}(r)$ may take a different form, but still will vanish much faster than $1/r$.)

Chapter 8 — Hamiltonian and Hamilton's equations of motion

MIDTERM REMARK

In polar coordinates

$$\frac{\partial L}{\partial \dot{\theta}} = p_{\theta} = \ell_z$$

$$L = L(r, \dot{r}, p_{\theta})$$

~~$\Downarrow$ Lagrange Eqn~~

[the two lines above are struck out by a large red X in the original]

$$L = T - V = L(q_1 \dots q_n, \dot{q}_1 \dots \dot{q}_n, t)$$

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \Big|_{\dot{q}_j, q_j} \right) - \frac{\partial L}{\partial q_i} \Big|_{\dot{q}_j, q_j}, \quad p_i = \frac{\partial L}{\partial \dot{q}_i} \Big|_{\dot{q}_j, q_j}.$$

Legendre transformation

$$q_1 \dots q_n, \dot{q}_1 \dots \dot{q}_n \longrightarrow q_1 \dots q_n, p_1 \dots p_n$$

$$f(x, y) \implies df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

In thermodynamics

$$\text{1st Law} \quad dU = \overset{[\text{fluid at constant } N]}{dQ} + dW = T dS - p dV, \quad U = U(S, V) \quad \text{internal energy}$$

We want T, V as variables.

$$F = A = U - TS \quad \text{Helmholtz free energy}$$

$$dA = dU - d(TS) = T dS - p dV - T dS - S dT = -S dT - p dV \implies A = A(T, V).$$

In general

The line below is written in the source with $f(x, y)$ on the left-hand side where df is meant.³

$$f(x, y) = u dx + v dy, \quad u = \frac{\partial f}{\partial x}, \quad v = \frac{\partial f}{\partial y}, \quad (x, y) \longrightarrow (u, y)$$

$$g = f - ux \implies dg = df - d(ux) = \cancel{u dx} + v dy - \cancel{u dx} - x du = -x du + v dy, \quad g = g(u, y).$$

The Hamiltonian

Let us look at $L(q_1 \dots q_n, \dot{q}_1 \dots \dot{q}_n)$, with

$$p_i = \left. \frac{\partial L}{\partial \dot{q}_i} \right|_{\dot{q}_j, q_j}$$

$$H = - \left(L - \sum_i p_i \dot{q}_i \right)$$

$$\begin{aligned} dH &= -dL + \sum_i d(p_i \dot{q}_i) \\ &= - \sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \cancel{\frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i} \right) - \frac{\partial L}{\partial t} + \sum_i (p_i \cancel{d\dot{q}_i} + \dot{q}_i dp_i) \end{aligned}$$

so that⁴

$$dH = -\frac{\partial L}{\partial t} + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i \right) \implies H = H(p_1 \dots p_n, q_1 \dots q_n, t).$$

$$\frac{\partial H}{\partial p_i} = \dot{q}_i, \quad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} \stackrel{\text{Lagrange Eqn}}{=} -\dot{p}_i, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.$$

CANONICAL EQS. OF MOTION

³Source reads “In general $f(x, y) = u dx + v dy$ ”; the left-hand side should be df , as the same page writes correctly two lines above ($f(x, y) \Rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$) and uses correctly one line below ($g = f - ux \Rightarrow dg = df - d(ux)$). Reproduced as written.

⁴The term $-\frac{\partial L}{\partial t}$ is inserted interlinearly in the source and carries no dt ; the same omission is repeated in the line below, where the first term of dH is written $-\frac{\partial L}{\partial t}$ rather than $-\frac{\partial L}{\partial t} dt$. Reproduced as written.